

Skills Summary

Finding and Verifying Inverse Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Inverting a linear rule

The method (works for every rule on this sheet): start from $x = f(y)$, then solve for y . Whatever you get is $f^{-1}(x)$.

Why it works: f sends inputs to outputs; the inverse asks the reverse question, so the old output x becomes the new input and the old input y becomes the unknown.

Example: Find the inverse rule for $f(x) = 4x + 9$.

Step 1. The rule multiplies by 4 and then adds 9, so the inverse must subtract 9 and then divide by 4 — write $x = f(y)$ and solve for y to get that automatically.

$$\begin{aligned}\text{Step 2. } x &= 4y + 9 \Rightarrow x - 9 = 4y \Rightarrow y = \frac{x - 9}{4} \\ \Rightarrow f^{-1}(x) &= \frac{x - 9}{4}\end{aligned}$$

Try it: Find the inverse rule for $h(x) = 6x - 5$.

$$\frac{9}{5+x} = (x)_{1-y} \text{ :check}$$

Skill 2 — Inverting a rational rule

Extra step: when y appears in a denominator, clear the fraction first, then gather *every* y term on one side and factor y out.

$$\text{Pattern: } x(\text{denominator in } y) = \text{numerator in } y \Rightarrow y \cdot (\dots) = (\dots) \Rightarrow y = \frac{\dots}{\dots}$$

Example: Find the inverse rule for $g(x) = \frac{3x}{x+2}$.

Step 1. The unknown sits in both the numerator and the denominator, so no amount of peeling operations will work — clear the fraction and factor y out instead.

$$\begin{aligned}\text{Step 2. } x(y+2) &= 3y \Rightarrow xy + 2x = 3y \Rightarrow xy - 3y = -2x \Rightarrow y(x-3) = -2x \\ \Rightarrow g^{-1}(x) &= \frac{2x}{3-x}\end{aligned}$$

Try it: Find the inverse rule for $k(x) = \frac{x-1}{x+5}$.

$$\frac{x-1}{1+x5} = (x)_{1-y} \text{ :check}$$

(!) Watch out: $\frac{3x}{x+2}$ does *not* invert to $\frac{x+2}{3x}$. The reciprocal of a rule is not its inverse.

Skill 3 — Inverting roots and powers

Undo a square root by squaring; undo a square by taking a root. For $f(x) = \sqrt{x-h} + k$ the inverse is $(x-k)^2 + h$; for $f(x) = (x-h)^3 + k$ it is $\sqrt[3]{x-k} + h$.

One-to-one first: an even power needs a restricted domain (such as $x \geq h$) before it has an inverse; an odd power never does.

Example: Find the inverse rule for $p(x) = \sqrt{x-5} + 2$.

Step 1. The last operation applied to the input is “add 2”, so the first thing the inverse must undo is that 2 — isolate the radical, then square.

Step 2. $x = \sqrt{y-5} + 2 \Rightarrow x-2 = \sqrt{y-5} \Rightarrow (x-2)^2 = y-5$
 $\Rightarrow p^{-1}(x) = (x-2)^2 + 5$

Try it: Find the inverse rule for $q(x) = \sqrt{x+3} - 1$.

check: $q^{-1}(q(x)) = (x+3) - 1 = x$

Skill 4 — Checking and using an inverse

The check: a rule f^{-1} is the inverse of f exactly when $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$. Simplify the composition; if it collapses to x , the rule is right.

Shortcut for a value: $f^{-1}(c)$ is the input that f sends to c , so you can also just solve $f(x) = c$.

Example: For $m(x) = 3x - 12$, check that $m^{-1}(x) = \frac{x+12}{3}$ and then find $m^{-1}(9)$.

Step 1. Two separate jobs: composition settles whether the rule is right, and substitution answers the numerical question — do the composition first so the value can be trusted.

Step 2. $m(m^{-1}(x)) = 3\left(\frac{x+12}{3}\right) - 12 = (x+12) - 12 = x \checkmark$

Step 3. $m^{-1}(9) = \frac{9+12}{3} = \frac{21}{3}$
 $\Rightarrow m^{-1}(9) = 7$

Try it: For $n(x) = 5x + 4$, find $n^{-1}(24)$.

check: $n^{-1}(n(x)) = \frac{5x+4-4}{5} = x$